



Gradient **G**

GD: flat vector

$$\text{vec}(\mathbf{G})$$

Frobenius norm

Muon: orthogonalize

$$\mathbf{U} \cdot \mathbf{V}^\top$$

(SVD, all $\sigma_i = 1$)

Spectral norm

Shampoo: precondition

$$\mathbf{L}^{-1/4} \mathbf{G} \mathbf{R}^{-1/4}$$

Kronecker curvature